100 Marks Question Paper

Part A: Multiple Choice Questions (MCQs) – 50 Marks

(2 marks each)

1. **Which of the following is an example of a supervised learning task?**
    - (a) Clustering
    - (b) Dimensionality reduction
    - (c) Regression
    - (d) Principal component analysis
2. **Which of the following metrics is used to evaluate a classification model?**
    - (a) R-squared
    - (b) Mean Squared Error (MSE)
    - (c) Accuracy
    - (d) Mean Absolute Error (MAE)
3. **Which algorithm finds the optimal weights by solving the equation θ=(XTX)−1XTy?**
    - (a) Stochastic Gradient Descent
    - (b) Ridge Regression
    - (c) Normal Equation
    - (d) Logistic Regression
4. **In Ridge Regression, which regularization term is added to the cost function?**
    - (a) L1 norm
    - (b) L2 norm
    - (c) Cross-entropy loss
    - (d) Softmax loss
5. **Which algorithm is used when the dataset has millions of features and we want to avoid the cost of inverting large matrices?**
    - (a) Normal Equation
    - (b) Polynomial Regression
    - (c) Gradient Descent
    - (d) SVD
6. **Which of the following regularization techniques can automatically select features by reducing some weights to zero?**
    - (a) Ridge Regression
    - (b) Lasso Regression
    - (c) Logistic Regression
    - (d) Softmax Regression
7. **Which gradient descent algorithm converges the fastest?**
    - (a) Batch Gradient Descent
    - (b) Stochastic Gradient Descent
    - (c) Mini-Batch Gradient Descent
    - (d) Normal Equation

8. **Which of the following is an example of a hyperparameter in a machine learning model?**
   - (a) Coefficients of the model
   - (b) Learning rate
   - (c) Input features
   - (d) Output labels
9. **What is the main purpose of cross-validation?**
   - (a) To increase training data size
   - (b) To tune hyperparameters
   - (c) To reduce bias in the training set
   - (d) To estimate how well the model generalizes to unseen data
10. **In the context of overfitting, which of the following techniques is NOT commonly used to reduce it?**
   - (a) Adding more features
   - (b) Reducing model complexity
   - (c) Using regularization
   - (d) Increasing the training data
11. **Which cost function is typically used in Logistic Regression?**
   - (a) Mean Squared Error
   - (b) Cross-Entropy Loss
   - (c) Hinge Loss
   - (d) L1 Loss
12. **In softmax regression, what is the output of the model?**
   - (a) Continuous values
   - (b) Binary classes
   - (c) Probabilities for each class
   - (d) Decision boundaries
13. **Which of the following is true for Polynomial Regression?**
   - (a) It adds regularization to the model.
   - (b) It introduces interaction terms and powers of features.
   - (c) It reduces the number of features.
   - (d) It only works for linear models.
14. **Which of the following methods is most affected by the scaling of features?**
   - (a) Logistic Regression
   - (b) Gradient Descent
   - (c) Normal Equation
   - (d) Ridge Regression
15. **Which of the following is a type of unsupervised learning?**
   - (a) Logistic Regression
   - (b) Support Vector Machines
   - (c) K-Means Clustering
   - (d) Polynomial Regression

16. **What is the main difference between Ridge and Lasso regression?**

- (a) Ridge uses L1 regularization, Lasso uses L2 regularization.
- (b) Ridge uses L2 regularization, Lasso uses L1 regularization.
- (c) Ridge reduces coefficients to zero, Lasso does not.
- (d) Lasso adds cross-entropy loss.

17. **Which of the following best describes the problem of underfitting?**

- (a) Model is too complex for the data
- (b) Model is too simple and cannot capture underlying patterns
- (c) Model has high variance
- (d) Model generalizes well to unseen data

18. **Which of the following is used to assess a model's generalization performance?**

- (a) Training error
- (b) Validation error
- (c) Learning rate
- (d) Coefficients of the model

19. **Which of the following is a characteristic of Stochastic Gradient Descent (SGD)?**

- (a) It converges to the exact minimum of the cost function.
- (b) It uses the whole dataset for each gradient step.
- (c) It updates parameters based on a single training instance at a time.
- (d) It is slower than Batch Gradient Descent.

20. **When tuning hyperparameters, which method randomly selects a combination of hyperparameters from a predefined grid?**

- (a) Random Search
- (b) Grid Search
- (c) Cross-Validation
- (d) Elastic Net

21. **Which of the following is NOT a type of gradient descent algorithm?**

- (a) Stochastic Gradient Descent
- (b) Batch Gradient Descent
- (c) Newton's Method
- (d) Mini-Batch Gradient Descent

22. **Which of the following describes hyperparameter tuning in machine learning?**

- (a) Tuning model parameters during training
- (b) Adjusting the parameters that control the learning process
- (c) Optimizing the final weights of the model
- (d) Reducing the loss function

23. **Which of the following is a key assumption in Linear Regression?**

- (a) Non-linearity of data
- (b) Independence of errors
- (c) High variance in model
- (d) Features must be scaled to [0, 1]

24. **Which type of regression is most likely to suffer from multicollinearity?**

- (a) Logistic Regression
- (b) Ridge Regression
- (c) Lasso Regression
- (d) Polynomial Regression

25. **In the context of softmax regression, which of the following is true?**
    - (a) It is a binary classification algorithm.
    - (b) It outputs the log-odds of a single class.
    - (c) It is used for multi-class classification problems.
    - (d) It applies L1 regularization to reduce features.

Part B: Short Answer Questions – 50 Marks

(5 marks each)

1. **Explain the difference between supervised learning and unsupervised learning. Provide examples for each.**
2. **What is overfitting? Mention two techniques to avoid it.**
3. **Describe the purpose of hyperparameter tuning and list two common hyperparameter tuning methods.**
4. **Write the cost function of Ridge Regression and explain the role of the regularization term.**
5. **How does Batch Gradient Descent differ from Stochastic Gradient Descent? Provide their advantages and disadvantages.**
6. **Explain the normal equation for Linear Regression and state one scenario where it may not be suitable.**
7. **What is Singular Value Decomposition (SVD)? How is it used in linear regression?**
8. **Define cross-entropy cost function used in softmax regression and explain why it's used.**
9. **What is the role of regularization in Lasso Regression? How does it help in feature selection?**
10. **Discuss the key differences between Logistic Regression and Linear Regression. When would you use each?**